compressed air to the atmosphere. When the pressure of the air in the reservoir reaches to 6.5 kg/cm^2, the governor valve automatically closes.

Reservoir tank

Reservoir tank is used for storing amount of air required for operating the brake system. It is made up of steel plate. There are provision given for air inlet and draining the impurities from the tank. In addition to these arrangement a safety valve is also located at the top of the tank for maintaining the air pressure.

Brake valve

This valve is fitted below the brake pedal. When the driver presses the brake pedal, the piston in the brake valve is moved downward. It connects the compressed air with the brake chambers. Therefore the compressed air is sent to the brake chamber located at the four wheels.

Brake chamber

Near to the each wheels, one brake chamber is located and connected with the wheels. The brake chamber is made up of steel plate. The brake chambers are separated by the diaphragm made up of rubber. One chamber is connected to the pressurized air coming from the brake valve and the another chamber is attached to the brake cam that operates the brake through the diaphragm. The diaphragm element is compressed by a spring and the spring is located in the brake chamber. While pressing the brake pedal, the compressed air against the spring force moves the diaphragm and tends to expand the brake shoes.

Slack adjuster

The slack adjuster is attached at the end of the diaphragm lever coming out from the brake chamber. The bottom edge of the slake adjuster is attached to the S cam which is used to operate the brake system. Therefore due to the compressed air diaphragm is moved forward and allowed to rotate the S cam that expands the brake shoes which actuate the brake.

Advantages of air brake system

1. As compared to mechanical and hydraulic brakes it has more power and efficiency. Hence it is mostly used in heavy vehicles.
2. The parts of the brake system can be fitted to anywhere in the vehicle. So the chassis structure becomes simpler.
3. Compressed air in the reservoir tank can be also used to operate air horn, screen wiper and to fill air for tyres.
4. The driver does not feel tiresome by operating the air brake because of its simpler actuation.
5. It has low maintenance cost.
6. The air brake system is widely used in heavy vehicle because it response is faster than any other systems and it works accurately.

6.4.3 Servo brake

For stopping the heavy vehicles, higher amount of braking force is required. Hence for increasing the braking force the driver has to apply more force on the brake pedal. During long distance travels the driver becomes tired due to the above problem. To avoid this difficulty and help the drivers any one type of brake is chosen and the brake system

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